TABLE OF CONTENTS

Page

2. PROJECT DESCRIPTION 2-1

2.1 Purposes and Project Objectives 2-1
2.2 Location, Site History, and Scope of the Project 2-1
2.3 Need for the Project 2-2
2.4 Benefits of the Project 2-2
2.5 Consideration of “With” and “Without” Project Scenarios 2-6
2.6 Consideration of Alignment Options 2-8
2.7 Construction Methodologies for Key Design Elements 2-16
2.8 Project Programme 2-20
2.9 Concurrent Projects 2-21
2.10 Key Public Concerns 2-29

List of Tables

Table 2.1	Summary of Alternative Mitigation Measures
Table 2.2	Alignment Options Comparison
Table 2.3	Summary of Possible Construction Methods for the Project
Table 2.4	Tentative Construction Programme
Table 2.5	Concurrent Projects
Table 2.6	Summary of Key Public Concerns and Follow-up Actions Taken

List of Figures

Figure 2.1	General Layout
Figure 2.1.1	General Layout (Sheet 1 of 4)
Figure 2.1.2	General Layout (Sheet 2 of 4)
Figure 2.1.3	General Layout (Sheet 3 of 4)
Figure 2.1.4	General Layout (Sheet 4 of 4)
Figure 2.2	Alignment Options
Figure 2.2.1	Alignment Options (Sheet 1 of 4)
Figure 2.2.2	Alignment Options (Sheet 2 of 4)
Figure 2.2.3	Alignment Options (Sheet 3 of 4)
Figure 2.2.4	Alignment Options (Sheet 4 of 4)
Figure 2.3	Location of Concurrent Projects

2. PROJECT DESCRIPTION

2.1 Purposes and Project Objectives

2.1.1.1 The Project's objective is to deliver a new road connecting Yuen Long Highway/ Tsing Long Highway and Fanling Highway, linking key new development nodes such as Ngau Tam Mei New Development Area (NTM NDA) and San Tin Technopole (STT).

2.2 Location, Site History, and Scope of the Project

2.2.1.1 The location of the Project is shown in **Figure 1.1**.

2.2.1.2 The alignment of the Project (San Tin Section) starts at the interchange of Yuen Long Highway / Tsing Long Highway on the west, and traverses the developed area in Mo Fan Heung, which are presently brownfield sites. It passes through the conservation areas near Fung Kat Heung and Ngau Tam Mei, goes underground at Ngau Tam Shan. The alignment travels along the villages near Saddle Pass, extends through the conservation areas near San Tin, continues underground at Hadden Hill (Ki Lun Shan), and finally connects to the Fanling Highway (Pak Shek Au Section).

2.2.1.3 Historically, the project site was predominantly rural and occupied by agricultural activities and village settlements. Currently, low-density residential development at Mo Fan Heung, Pok Wai, Kwu Tung, etc., villages such as Yau Tam Mei Tsuen, Ki Lun Tsuen, scattered brownfield operations, farmland/fishponds are found within or in the vicinity of the project.

2.2.1.4 The scope of the Project is to provide a carriageway connecting Yuen Long Highway/ Tsing Long Highway and Fanling Highway (Pak Shek Au Section). It comprises the following key elements, the general layout is shown in **Figure 2.1**.

(a) Construction of an approximately 9.0-kilometre (km) long dual three-lane carriageway, mainly in the form of at-grade road / tunnel / viaduct. In total, about 2.5 km is in the form of tunnel in mainline;

(b) Construction of five interchanges located at Yuen Long Highway/Tsing Long Highway, Ngau Tam Mei NDA, San Tin Technopole, Fanling Highway and Kwu Tung North NDA. This includes approximately 150m tunnel length at the interchange slip road connecting STT; and

(c) Associated junction modifications and slip roads required for the project, civil, geotechnical, landscape, road and drainage works, ancillary buildings, traffic control and surveillance system, toll collection facilities, electrical and mechanical installations, reprovisioning of facilities affected by the proposed works, environmental mitigation measures and administration building / kiosk, portal plant building/ plant room, etc.

2.3 Need for the Project

2.3.1.1 With the progressive development of STT and NTM NDA, traffic congestion is anticipated at San Tin Highway during peak hours in 2036. The Project will alleviate the San Tin Highway traffic congestion by providing an alternative route.

2.3.1.2 In longer term, the transport demand for east-west connectivity within the Northern Metropolis and cross-boundary connections is projected to increase substantially – driven by the progressive development of the Northern Metropolis and the gradual increase on the residential and employment population. The Project, together with the other three sections of NMH, can provide alternative route from New Territories East and West, and alleviate the traffic congestion on Yuen Long Highway, San Tin Highway and Fanling Highway.

2.4 Benefits and Alternative Mitigation Measures of the Project

2.4.1.1 The Project can bring the following benefits:

Environmental Benefits and Environmentally Friendly Design

(a) Avoidance of direct impact to ecologically sensitive areas - The recommended alignment and its works footprint have been thoroughly considered and evaluated from alternative alignment options proposed in previous stage studies (**Section 2.6** refers). This process has resulted in the recommended alignment option, which aims to avoid both the direct and indirect potential impacts on recognised sites of conservation importance and other ecologically sensitive areas, such as the Lam Tsuen Country Park (LTCP), the proposed Nam Sang Wai Wetland Conservation Park and natural watercourses at Ngau Tam Mei. The locations of piers are also carefully designed to be outside the Ngau Tam Mei natural watercourses.

(b) Wetland Habitat Enhancement through Integration and Expansion – Under the original alignment, an interchange, I/C-03, (with associated site office and ancillary facilities) was proposed to the west of Ngau Tam Mei Water Treatment Works (NTMWTW), where scattered and fragmented wetland habitats are currently found. Compared to the original alignment, the recommended alignment has been refined, with the Interchange I/C-03 relocated to the south of NTM NDA (refer to **Figure 2.1.2**). The viaduct level has also been raised as much as practicable to provide greater clearance allowing natural sunlight to reach the habitats underneath the viaduct structures. The relocation of Interchange I/C-03 to the south of NTM NDA will also provide an opportunity to integrate and expand the wetland habitats to the west of NTMWTW (refer to **Figure 2.1.3**). Furthermore, the mainline will be constructed as an elevated viaduct with a limited number of piers, spanning over the wetland habitats to reduce direct ecological impact to this wetland habitats. Taking advantage of the Project implementation, the area will not only be reinstated, but the existing fragmented wetland habitats will also be integrated and enhanced into a connected wetland habitat. The total wetland area will increase from approximately 3 hectares of fragmented ponds to more than 5 hectares of integrated wetland habitats through the incorporation of adjacent village / orchard land. With improved design and management, the enhanced wetland habitat will support higher biodiversity, greater ecological

value and increased carrying capacity. It will also offer a suitable foraging environment for various relevant species, including waterbirds.

(c) Allowing Wildlife Movement by Adopting Intermittent At-grade and Viaduct Design – A section of the alignment runs adjacent to the woodland habitat at the edge of the LTCP near Long Ha (refer to **Figure 2.1.2**). The alignment profile has been raised as far as practicable, taking into account site constraints such as CLP cables, pylons and traffic design requirements. The design has been optimised to avoid encroachment into LTCP while maximising the length of the viaduct section. Each of the at-grade sections is no greater than ~350 m so as to avoid creating potential hurdles to wildlife movement. The space beneath the viaduct sections provides an open corridor to facilitate wildlife movement to and from LTCP. For the at-grade sections, retaining wall on the slope would be constructed to avoid wildlife roadkill.

(d) Viaduct Level Arrangement and Carriageway Separation for Enhancing Sunlight Penetration and Ecological Benefits – The Project includes slip roads crossing the Kam Tin River (refer to **Figure 2.1.1**), where the viaduct level has been adjusted to provide headroom of about 10~15m and minimise shading effects on the water body and associated habitats. In the vicinity of NTMWTW, to accommodate the required clearance from the existing CLP cable, the headroom beneath the viaduct is approximately 10m for the northbound and ranges from 5m to 10m for the southbound. The dual carriageways are separated before entering the tunnel portal (refer to **Figure 2.1.3**). The resulting gap between the carriageways enables natural sunlight to reach the ground below and is not expected to cause adverse ecological impacts. Furthermore, the positioning of piers have been refined to avoid encroachment into natural watercourses. This approach helps to minimize potential water quality and ecological impact.

(e) Adoption of Non-dredged Method for Construction Works at Kam Tin River - The significance of ecological value of Kam Tin River is well recognised. Given the large span of the viaduct crossing Kam Tin River, the pile /pier support within the river is inevitable (refer to **Figure 2.1.1**). The construction of viaduct pier/ pile will adopt the non-dredging method. For example, if bored pile construction is adopted, steel casing could be adopted to isolate the surrounding riverbed. The material within the steel casing will be excavated within the casing for placing reinforcement and concrete of the pile. This minimises the disturbance to the river.

(f) Synergistic Woodland Enhancement near She Leng – A Woodland compensation of approximately 17 ha is proposed near She Leng, which is currently characterised by grassland or shrubland of low or low to moderate ecological value. The site is also adjacent to proposed woodland compensation areas under other projects, including Lok Ma Chau Loop, STT and Development of the Loop – Eastern Connection Road (ECR). Woodland compensation at this location will integrate the overall compensatory woodland area and improve habitat quality and ecological connectivity with adjacent woodland habitats.

(g) Minimisation of slope cutting through adoption of elevated alignment – Given the hilly terrain to the south of Ngau Tam Mei New Development Area (NTM NDA), the road alignment has been elevated (refer to **Figure 2.1.2**). This

adjustment not only minimises the scale of site formation works, but also takes into account other constraints, such as the clearance requirement for CLP power cables. Additionally, a larger headroom would provide greater clearance between the structure and the habitats so as to allow natural sunlight to shine on the habitats underneath the viaduct structures.

(h) Minimisation of slope cutting through soft ground tunneling at portals – When the alignment passes through Ngau Tam Shan and Hadden Hill, soft ground tunnelling has been adopted at portals. Specifically, before the tunnel excavation reaches bedrock, soft soil stratum will be improved with injection grouting around the tunnel profile. Temporary support such as pipe roofing and steel lattice girder/rib will be installed during the tunnel excavation by mechanical method. With the adoption of soft ground tunnelling, the extent of site formation is significantly reduced, which helps retain a larger area of natural terrain and minimise the generation of construction and demolition (C&D) materials.

(i) Minimisation of slope cutting through construction of retaining walls and soil nailing – For sections where extensive cut slopes would otherwise be required, retaining walls / soil nails or combination of both may be proposed. This measure aims to reduce the extent of slope excavation, allow for steeper slope gradients, and thereby minimise the project's environmental footprint, as well as acting as a barrier to prevent wildlife roadkill.

(j) Avoidance of direct impacts on Sites of Archaeological Interest (SAI) – Avoid any excavation works which will cause potential disturbance to the soil matrix within the boundary of Ngau Tam Mei SAI to avoid potential direct impacts on the SAI (refer to **Figure 2.1.2**).

(k) Adopt Drill and Blast (D&B) method for tunnel construction – Adopt D&B method over TBM (Tunnel Boring Machine) method for tunnel construction to minimize the generation of C&D materials as much as practicable.

(l) Diverting the existing traffic flow between northeast and northwest of the New Territories, lowering vehicular emissions from idling or congested traffic and shortening travel times and results in further reducing fuel consumption, alleviating potential air quality and noise impact during operation phase.

Social Benefits

(m) Serving the traffic demand for better east-west connectivity within the Northern Metropolis and cross-boundary connections.

(n) Providing more connectivity to the existing road and the development nodes in the Northern Metropolis, which improves transport efficiency and enhances connectivity.

(o) Encouraging and enabling the public to easily access various land-based boundary control points (BCPs) via major roads and railways, boosting the connectivity and synergy effect among these BCPs as well as the benefits of the cross-boundary infrastructure.

(p) Supporting 'infrastructure-led' growth at the development nodes and creating job opportunities.

2.4.1.2 Due consideration has been given above to address the environmental challenges encountered in the design of the Project. Environmental protection will be prioritised and fulfilled to the greatest extent possible. Various key actions have been implemented to address the environmental challenges and summarised in **Table 2.1**.

Table 2.1 Summary of Alternative Mitigation Measures

Alternative Mitigation Measures	Details	Environmental Benefits	Environmental / Engineering Disbenefits
Avoidance of Recognised Sites of Conservation Importance and Other Ecologically Sensitive Areas	• Avoid both the direct and indirect potential impacts on recognised sites of conservation importance and other ecologically sensitive areas, such as the LTCP, the proposed Nam Sang Wai Wetland Conservation Park and natural watercourses at Ngau Tam Mei • Relocate I/C 03 to the south of NTM NDA	• Avoid direct ecological impact on the recognised sites of conservation importance and other ecologically sensitive areas	• More interface with existing CLP, and Express Rail Link (XRL) at the relocated I/C-03
Avoidance of Ngau Tam Mei SAI	• Avoid any excavation works which will cause potential disturbance to the soil matrix within the boundary of Ngau Tam Mei SAI	• Avoid archaeological impact on Ngau Tam Mei SAI	• Nil
Minimisation of C&D materials	• Minimisation of slope cutting through adoption of elevated alignment to the south of NTM NDA	• Minimise the extent of slope cutting required and thus minimising the	• Nil

Alternative Mitigation Measures	Details	Environmental Benefits	Environmental / Engineering Disbenefits
	• Minimisation of slope cutting through soft ground tunnelling at portals of Ngau Tam Shan and Hadden Hill • Minimisation of slope cutting through construction of retaining wall and soil nailing to allow for steeper slope gradient and reduce extent of excavation • Adoption of D&B method for tunnel construction over TBM method	generation of C&D materials • Minimise C&D materials generated during tunnel construction	
Minimisation of water quality and ecological impact	• Adopt non-dredged method for construction works at Kam Tin River	• Minimise the disturbance to the water flow regime of Kam Tin River and ecological impact	• Less flexibility on construction methods, leading to associated cost and programme implication
Minimisation of ecological impact	• Adopt viaduct with limited number of piers at the wetland habitat to the west of NTMWTW • Adopt intermittent at-grade and viaduct design for the alignment near LTCP • Adjust the viaduct level of the alignment to provide sufficient headroom	• Minimise habitat loss due to piers • Minimise potential ecological impact due to shading effect of the viaduct • Facilitate wildlife movement to and from LTCP	• By maximising the elevated viaduct section and headroom, clearance with overhead CLP cable is reduced and leading to construction constraint

2.5 Consideration of "With" and "Without" Project Scenarios

2.5.1 Scenario "with" Project

2.5.1.1 Upon the implementation, the Project will provide an alternative route from Yuen Long Highway / Tsing Long Highway to Fanling Highway which will alleviate the traffic pressure on existing road network with progressively increasing traffic demand.

2.5.1.2 It will also link up the development nodes with provision of direct access to areas such as NTM NDA, STT and Kwu Tung North New Development Area (KTN NDA) to significantly improve transport efficiency of the strategic road network. The enhanced accessibility will in turn drive and expedite the implementation of

development projects within these areas and fully unleashes the development potential.

2.5.1.3 The public and visitors could make use of the developed major road network to conveniently travel to the various land-based boundary control points in the Northern Metropolis. It will effectively improve the cross-boundary road connection for passengers and goods in Hong Kong.

2.5.1.4 The Project would constitute one of the four sections of NMH which effectively connects the New Territories East and West. The driving distances within the Northern Metropolis will be greatly reduced after the commissioning of the NMH. The reduced traffic time would save the energy consumption and the improved traffic efficiency would minimise potential air quality impact and noise impact to the sensitive receivers.

2.5.1.5 In addition, the Project incorporates ecological-friendly design measures, which help prevent wildlife from entering carriageways and reduce roadkill risk. The Project also enables enhanced ecological connectivity through planned habitat creation, such as wetland compensation areas that complement existing ecological resources at the area to the west of the NTMWTW, thereby improving long-term habitat continuity and supporting biodiversity.

2.5.2 Scenario "without" Project

2.5.2.1 Without the implementation of the Project, traffic congestion would be expected at road section of San Tin Highway in year 2036 during peak hour with the progressive development of development nodes such as STT and the NTM NDA.

2.5.2.2 The connectivity of development nodes will not be as direct and convenient as with the Project which in turn limits the potential of development area.

2.5.2.3 The overstress of existing road network would cause traffic congestion which in turn induces potential air quality and noise impacts to the sensitive receivers.

2.5.2.4 Without the Project, the east-west traffic connection in Northern Metropolis would heavily rely on the existing San Tin Highway which will operate beyond its capacity.

2.6 Consideration of Alignment Options

2.6.1 General

2.6.1.1 The alignment under the Project consists of 5 interchanges (I/C-02 to I/C-06) from Yuen Long Highway / Tsing Long Highway to Fanling Highway. Interim interchanges (I/C-03 to I/C-05) are provided to connect some development nodes and existing roads. The I/C-02 and I/C-06 will reserve stub ends for connection to the future Tin Shui Wai section and Kwu Tung section of NMH respectively.

(a) I/C-02 will reserve stub end at west for future connection to Tin Shui Wai Section of NMH.

(b) I/C-02 to I/C-03 connects Yuen Long Highway/ Tsing Long Highway and NTM NDA

(c) I/C-03 to I/C-04 connects NTM NDA and STT

(d) I/C-04 to I/C-05 connects STT and Fanling Highway

(e) I/C-05 to I/C-06 connects Fanling Highway and Kwu Tung North NDA

(f) I/C-06 will reserve stub end at north for future connection to Kwu Tung Section of NMH

2.6.1.2 Original Alignment Option was initially derived from the SSMR FS under Agreement No. CE7/2018(TT).

2.6.1.3 After commencement of the Investigation stage, more detailed analysis has been conducted, and several alignment options have been further explored aiming to optimise the scheme and to enhance environmental benefits.

2.6.1.4 With preliminary review and screening, three alignment options (inclusive of the Original Alignment Option) have been shortlisted and taken forward for evaluation from perspectives of site constraints, traffic performance, engineering constructability, social acceptance, environmentally friendliness, etc to suggest the recommended alignment.

2.6.1.5 Alignment Option 3 is recommended after robust considerations. **Figure 2.2 (Figure 2.2.1-2.2.4)** delineates the three alignment options. Sections below present the considerations of alignment options with elaboration.

2.6.2 Alignment Option 1 (Original Alignment)

2.6.2.1 The original alignment is primarily designed as an underground scheme to avoid encroachment upon above-ground constraints.

I/C-02 to I/C-03

2.6.2.2 The mainline of alignment would be constructed in the form of tunnel from Nam Sang Wai Wetland and across the Kam Tin River from future Tin Shui Wai Section of NMH at west (refer to **Figure 2.2.1**). In view of the geological condition, Tunnel Boring Machine (TBM) tunnels, supported by a shaft each at two ends for the launching and retrieving TBM, is proposed for the underground section at the west of LTCP before entering the hill. Both shafts will be constructed using the open cut method—this involves digging deep, large-scale excavation and installing lateral support structures to keep the excavation stable. After entering the LTCP where the terrain is hilly, tunnelling will be carried out primarily using the drill-and-blast method, with localised supplementary mechanical breaking where necessary.

2.6.2.3 To provide the connection at I/C-02 to existing San Tin Highway while catering the level difference, depressed slip roads are required to connect the existing at-grade San Tin Highway near Mo Fang Heung and tunnel mainline. The alignment mainline keeps the tunnel form and traverses the LTCP in parallel to existing water tunnel at south. The mainline tunnel would go under the existing Ngau Tam Mei Water Reservoir and the existing underground Express Rail Link (XRL) (refer to **Figure 2.2.2**).

2.6.2.4 Before arriving at the existing NTMWTW, the alignment curves towards north and become day light section given the topographical feature. The mainline and I/C-03 would be accommodated between the narrow strip of land between the existing NTMWTW and the future NTM NDA (refer to **Figure 2.2.3**).

I/C-03 to I/C-04

2.6.2.5 After I/C-03 and the NTMWTW, the alignment curves toward east to across the Ngau Tam Shan in tunnel form (refer to **Figure 2.2.3**) and then bends north subsequently toward the next connection I/C-04 which would become day light section connecting the STT. The I/C-04 location is considerately selected at the low point between Ngau Tam Shan and Hadden Hill (refer to **Figure 2.2.4**) to minimise the site formation while not encroaching the STT. This inevitably creates a double curve tunnel layout between I/C-03 and I/C-04.

I/C-04 to I/C-05

2.6.2.6 After the I/C-04, the alignment encounters the Hadden Hill and changes to tunnel form in parallel to the existing water tunnel. At the west side of existing Kwu Tung Reservoir, there is small section of valley which requires the alignment to span over in viaduct form. After passing the Hadden Hill, the alignment becomes day light section at I/C-05 to provide connection to Fanling Highway in viaduct form (refer to **Figure 2.2.4**).

I/C-05 to I/C-06

2.6.2.7 After I/C-05, the alignment flyovers the existing Fanling Highway (refer to **Figure 2.2.4**). Considerations shall be given to the headroom between the alignment and the existing elevated interchange. Location of I/C-05 and I/C-06 may appear in close distance, however they are serving different traffic catchment and considered necessary from traffic point of view. While I/C-05 connects to the Fanling Highway, I/C-06 offers opportunities for the KTN NDA currently under construction which will boost the development of the new development area.

2.6.3 Alignment Option 2

2.6.3.1 The alignment option aims to explore opportunity to shortening the alignment length while avoiding the double curve between I/C-03 and I/C-04 to improve the traffic performance.

I/C-02 to I/C-03

2.6.3.2 At I/C-02, the mainline of alignment starts near Wing Kei Tsuen with slip roads connecting San Tin Highway (refer to **Figure 2.2.1**). After a short section of tunnel going under the LTCP, the alignment would curve north earlier than Alignment Option 1 to approach the south of NTM NDA and provide connection to NTM NDA as I/C-03 in a day light section (refer to **Figure 2.2.2**). The existing rows of CLP pylon at south (132kV and 400 kV respectively) impose spatial constraints to the alignment and interchange arrangement. The interchange would inevitably encroach upon the NTM NDA.

I/C-03 to I/C-04

2.6.3.3 After I/C-03, differing from Alignment Option 1, the alignment runs at the south of the NTMWTW and then turns toward north (refer to **Figure 2.2.3**). This arrangement attempts to avoid creating the double curve alignment layout between I/C-03 and I/C-04 as Alignment Options 1 and 3. The alignment at the south of NTMWTW is a day light section given the topography feature.

2.6.3.4 However, in view of the site constraints (e.g. the NTMWTW, the CLP pylons and the topography condition) and the limitation by prevailing traffic design requirement (e.g. road gradient and radius), it is envisaged that the site formation, piling work for viaduct at the south of NTMWTW, and tunnel portal formation at the south-east of NTMWTW would be carried out in the Conservation Area when the alignment enters the Hadden Hill .

2.6.3.5 At I/C-04, the alignment provides connection to the STT (refer to **Figure 2.2.4**). Since this alignment option aims to shorten the overall length, the alignment mainline is shifted west compared to the Alignment Option 1. The I/C-04 would inevitably encroach the STT. The mainline tunnel would be closer to the existing water tunnel compared to the Alignment Option 1. The tunnel portal formation near I/C-04 would encroach more area of the Conservation Area at the north and south of I/C-04 compared to Alignment Option 1.

I/C-04 to I/C-05

2.6.3.6 After I/C-04, the alignment passes through Hadden Hill in the form of tunnel (refer to **Figure 2.2.4**). The alignment is further away from the existing Kwu Tung Reservoir.

2.6.3.7 After passing through Hadden Hill, the alignment becomes day light section. The tunnel portal formation would encroach the Conservation Area at north of Hadden Hill.

I/C-05 to I/C-06

2.6.3.8 The alignment joins in Fanling Highway in north-east direction. At I/C-05, the alignment runs towards the north and form I/C-06 (refer to **Figure 2.2.4**).

2.6.4 Alignment Option 3 (Recommended Alignment)

2.6.4.1 This alignment option aims to explore the opportunity of minimising the tunnel length while improving traffic connectivity at I/C-02.

I/C-02 to I/C-03

2.6.4.2 The location of I/C-02 shifts south to the existing interchange of Yuen Long Highway/ Tsing Long Highway in the form of viaduct (refer to **Figure 2.2.1**). Slip road diverges from the mainline not only reserve connectivity to the San Tin Highway (provided by NMH Tin Shui Wai Section), but it also provides additional connectivity to the existing Yuen Long Highway / Tsing Long Highway (Route 3) by crossing over the Kam Tin River toward west. Given the site constraints, the level of alignment is optimised as practicable to minimise the impact to the flight path of birds while meeting highway clearance and headroom requirements. The piers for the viaduct within Kam Tin River would be arranged to in similar pattern as the existing Yuen Long Highway to minimise the hydraulic impact. Construction method of pile within Kam Tin River would be non-dredging method to minimise the disturbance to the riverbed.

2.6.4.3 At east side of Kam Tin River, the alignment mainline will fly over the at grade San Tin Highway and run in parallel to the San Tin Highway in form of viaduct. Modification work of existing San Tin Highway is required for accommodation of the proposed piers. Compared to Alignment Options 1 and 2, which have substantial impact on the existing San Tin Road and San Tam Road, the elevated viaduct scheme in Alignment Option 3 provides flexibility in the pier arrangement to minimise impact on the existing road and their utilities underneath, especially the high-pressure gas mains under the San Tam Road.

2.6.4.4 The alignment near I/C-02 would also be along the edge of the proposed wetland compensation area adjacent to Sha Po Village for the Ngau Tam Mei NDA. Recognising the environmental significance of the wetland compensation, the alignment has been shifted westward, and the interchange footprint has been minimised. Despite various constraints including the existing interchange, the existing CLP cable and pylons, and the nearby Conservation Area, etc.), the alignment has been carefully adjusted to run along the edge of the proposed wetland compensation area. This avoids potential fragmentation of the proposed wetland compensation area. Furthermore, ongoing close liaison with interface

project team will be maintained to minimise the impact during the construction phase.

2.6.4.5 Until the alignment reaching Long Ha, the alignment curves toward east and remains as day light section (refer to **Figure 2.2.1**). The CLP cable imposes limitation on the road level at the turning location. To minimize site formation while managing the large elevation difference between the road and the adjacent ground, a soil-nailed retaining wall will be built. Its underground soil nails will extend into LTCP's subsurface—this design effectively prevents any above-ground construction from encroaching on the LTCP. Current alignment also manoeuvres East of San Tam Road avoiding existing CLP Cable towers and planned NOL facilities whilst locating at the edge of Kai Kung Shan near Pok Wai Village area to minimize site formation, reduce impact to those mentioned facilities and land requirements. In between the NTM NDA and the LTCP, the alignment runs along the hillside to minimise the site formation and land requirements. The alignment then connects to south of NTM NDA at I/C-03 (refer to **Figure 2.2.2**). Compared to Alignment Option 1, the relocation of Interchange I/C-03 from the NTMWTW to the southern part of the NTM NDA not only provides more direct connectivity to the NTM NDA, but also minimises the footprint on the fishpond near the NTMWTW, thereby reducing environmental impact.

I/C-03 to I/C-04

2.6.4.6 After I/C-03, the alignment continues to run along the south of NTM NDA, before arriving the existing NTMWTW, the alignment curves towards the north.

2.6.4.7 At the tunnel portal near NTMWTW, the separation between the northbound and southbound tunnel tubes has been reviewed and narrowed compared to Alignment Option 1 (refer to **Figure 2.2.3**). This helps reduce the length of the cross-passages, which connect the two tunnel tubes at regular intervals. Site formation for the tunnel portal can also be minimised through the reduced tunnel footprint.

2.6.4.8 After passing through Ngau Tam Shan, the tunnel section of the alignment has been shifted slightly eastward compared to Alignment Option 1 (refer to **Figure 2.2.4**), to an area with lower topography. This enables shortening of the tunnel length and minimisation of site formation.

I/C-04 to I/C-05

2.6.4.9 Near I/C-04, the interchange has been fine-tuned from a free-flow junction to a signalised junction while maintaining satisfactory junction performance relative to Alignment Option 1 (refer to **Figure 2.2.4**). This improvement helps minimise the interchange footprint, which in turn reduces cost, land take, and environmental impact.

I/C-05 to I/C-06

2.6.4.10 Near Interchange I/C-05, the alignment mainline has been shifted slightly eastward to avoid potential interface with the STT near Fanling Highway. I/C-06 has also been fine-tuned from a free-flow junction to a signalised junction while maintaining satisfactory junction performance, resulting in savings in cost and land take compared to Alignment Option 1 (refer to **Figure 2.2.4**).

2.6.5 Alignment Options Comparison

2.6.5.1 A comparison of the alignment options from engineering, traffic, environmental and other perspectives is presented below.

Table 2.2 Alignment Options Comparison

Alignment Option 1 (Original Alignment)	Alignment Option 2	Alignment Option 3 (Recommended Alignment)
Pros ✓ • Mostly tunnel and minimise interface with aboveground facility (e.g.CLP Pylon). • Minimise ecological impact by tunnel form near I/C-02.	Pros ✓ • Mostly tunnel and minimise interface with aboveground facility (e.g.CLP Pylon). • Minimum alignment length. • Minimise ecological impact by tunnel form near I/C-02.	Pros ✓ • Better traffic connectivity with direct connection to Route 3 via Tsing Long Highway. • Mostly aboveground and minimise interface with underground facility (e.g. Guangzhou-Shenzhen-Hong Kong Express Rail Link (XRL), existing water tunnel). • Lane changing and weaving flexibilities are allowed. • Avoid direct ecological impact on the wetland habitat near Nam Sang Wai. • Completely avoids the LTCP. Minimise ecological impact. • Lesser the construction and demolition materials when compare with Option 1 & 2, due to reduction in tunnel length. • Lower capital and re-current cost with shorter tunnels. • Direct connection to the planned residential sites of NTM NDA at I/C-03. • Signalised junctions at I/C-04 and I/C-06 minimise interchange footprint while in turn reduce cost, land take and environmental impact.

Alignment Option 1 (Original Alignment)	**Alignment Option 2**	**Alignment Option 3 (Recommended Alignment)**
Cons ✕ • I/C-02 depressed slip road and tunnel work construction have significant impact on San Tin Highway, Castle Peak Road, and San Tam Road. • Tunnel Boring Machine (TBM) Method near I/C-02 before entering LTCP. High risk of tunnelling in shallow ground cover which may induce ground settlement and affect the population nearby and heavy traffic on San Tin Highway. • Temporary deep and large area excavation in wetland habitat at Nam Sang Wai for deep launching/retrieval shaft for TBM launching which involve dredging and dewatering. Launching the TBM from the wetland at Nam Sang Wai would cause direct ecological impact to the wetland area. • Large extent of tunnel parallel run and close to water tunnel which may have cost and programme implications. • Lane change and weaving opportunities are limited due to tunnel portions. • No direct connection to Tsing Long Highway. The connection to Tsing Long Highway will have to via existing San Tin Highway. • Generate more construction and demolition materials due to long tunnel.	Cons ✕ • I/C-02 depressed slip road and tunnel work construction have significant impact on San Tin Highway, Castle Peak Road, and San Tam Road. • TBM Method near I/C-02 before entering LTCP. High risk of tunnelling in shallow ground cover which may induce ground settlement and affect the population nearby and heavy traffic on San Tin Highway. • Temporary deep and large area excavation in wetland habitat at Nam Sang Wai for deep launching/retrieval shaft for TBM launching which involve dredging and dewatering. Launching the TBM from the wetland at Nam Sang Wai would cause direct ecological impact to the wetland area. • Large extent of tunnel parallel run and close to water tunnel which may have cost and programme implications. • Lane change and weaving opportunities are limited due to tunnel portions. • No direct connection to Tsing Long Highway. The connection to Tsing Long Highway will have to via existing San Tin Highway. • Generate more construction and demolition materials due to long tunnel.	Cons ✕ • Slightly longer alignment length and thus a very slight increase in driving time. • Potential impact on flight path near I/C-02.

Alignment Option 1 (Original Alignment)	Alignment Option 2	Alignment Option 3 (Recommended Alignment)
• High capital and re-current cost with more tunnels. • Larger interchange footprint at I/C-04 and I/C-06.	• High capital and re-current cost with more tunnels. • Affects the land use planning of the NTM NDA and the STT. • Larger interchange footprint at I/C-04 and I/C-06.	

2.6.5.2 After comparison, the Alignment Option 3 is recommended, as it delivers improved constructability, minimises capital and recurrent costs, minimised impact on ecologically sensitive areas, and meets the required traffic performance.

2.7 Construction Methodologies for Key Design Elements

2.7.1 Key Design Elements

2.7.1.1 The key construction elements of the Project include construction of at-grade roads, viaduct, tunnel, administration, portal plant buildings / plant room and satellite control buildings, retaining wall / slope strengthening.

2.7.1.2 The selection of construction methodologies has been carefully considered to minimise environmental impacts, optimise constructability, and reduce construction duration.

2.7.1.3 Possible construction methods which have been investigated for the construction of the Project and the reasons for selection are summarised in **Table 2.3** below and detail description are presented in the following paragraph.

Table 2.3 Summary of Possible Construction Methods for the Project

Construction element	Possible construction method	Reason
At-grade roads	Cut and fill method	Due to the hilly terrain, the cut and fill method is proposed to cope with the level difference. The cut material from higher terrain will be used to fill lower-lying and achieve cut-and-fill balance.
Viaduct	Precast prestressed concrete construction method	Better control of quality and workmanship for works in fabrication yard. This also reduces the generation of construction and demolition (C&D) materials on site and minimises construction air quality and noise impact
Tunnel	Drill-and-blast	Adopting D&B method can avoid excessive over-excavation and C&D materials. Small section with Mechanical Break will be adopted where poor ground conditions or sensitive receivers are encountered. TBM is considered not preferrable considering the cost, constructability, programme, environmental impact.
Construction works at Kam Tin River	Provision of temporary steel decking and non-dredged piling construction method with staged construction	To minimise the ecological impact and water quality impact at Kam Tin River.
Construction of Administration, Portal Plant Building / Plant Room and Satellite Control Buildings	Conventional reinforced concrete construction method	The buildings are small in scale and low rise. The construction method suits for site in low accessibility. Precast concrete elements would be explored wherever possible to minimise

Construction element	Possible construction method	Reason
		on-site work and the associated environmental impacts, including the generation of C&D materials, construction air quality and noise impact, and other disturbance to surrounding environment.
Construction of retaining wall / slope strengthening	Conventional reinforced concrete retaining wall construction, tie back and slope strengthening by soil nail	The construction method suits the stage by stage slope cutting and site formation works, and minimise the extent of slope formation.

2.7.2 Construction of At-grade Roads

2.7.2.1 Cut and fill method is a widely adopted method and would be proposed to cope with the level difference for the formation of at-grade road. The excavated material from areas where the terrain is higher than the road level will be used to fill lower-lying areas, effectively achieving a cut-and-fill balance. For areas with generally low ground levels, the excavated material will be used to fill embankments for road leveling, minimising the disposal of construction and demolition materials. In general, slopes will be formed on both sides of the road where space permits. However, if constrained by facilities or ecological sensitive areas, retaining structures will be used to minimise the footprint and avoid encroachment, ensuring the project complies with environmental requirements.

2.7.3 Construction of Viaduct

2.7.3.1 To minimise the on-site work and associated environmental impact, precast prestressed concrete construction method would be employed in general. The precast concrete girder would be manufactured in casting yard offsite and transported to site for final positioning and stitching. However, under certain restrictive conditions where the precast construction method is not feasible (e.g. when large-span arrangement require large precast concrete components or when site accessibility is limited for delivery of precast component) the in-situ cast construction method will be used instead.

2.7.4 Construction works at Kam Tin River

2.7.4.1 The piers for the viaduct within Kam Tin River would be arranged to in similar pattern as the existing Yuen Long Highway to minimise the hydraulic impact. Temporary decking would generally be required for the mobilisation of piling machine. Temporary support will be installed together with steel decking to minimise the temporary footprint and occupation of the riverbed. To further reduce disturbance to the watercourse, staged construction will be adopted — for example, temporary steel decking and associated viaduct piles will be constructed on one side first; upon completion, these temporary decking will be dismantled, followed by construction of temporary deck and viaduct piles on the opposite side. The construction method of pile within Kam Tin River would be non-dredging method. Steel casing piles will first be installed to founding level to create a sealed working enclosure. Materials within the casing will then be excavated and removed. A tremie

pipe will be deployed to place concrete at the base of the casing, forming a secure base seal and minimising potential leakage. Steel reinforcement cages will be prefabricated off-site to reduce on-site works, then lowered into the casing prior to concrete pouring.

2.7.5 Construction of Tunnels

2.7.5.1 Based on the available geological profile, it is envisaged that tunnel excavation will be mostly carried out in good quality volcanic rock masses. They are typically excavated by drill & blast (D&B) method. Other forms of excavation such as mechanical and chemical splitting are not cost effective except for either small volume of excavation or at locations where blasting would pose too great nuisance or be hazard (e.g. in close proximity to sensitive receivers). When encountering softground (e.g. soil layer) or low cover conditions, mechanical excavation with robust temporary support would be adopted.

2.7.5.2 TBM has been assessed and considered inappropriate for adoption for the Project, based on the considerations set out below.

- The largest hard-rock TBM available globally is approximately 15 m in diameter, which is below 16m to 19m span required for 3-lane tunnels with sightline widening in the Project.

- TBM involves relatively high initial capital investment and longer mobilisation periods, particularly associated with the setting of launching and receiving areas and machine mobilisation.

- TBMs inherently create circular cross-sections in fixed dimensions, which does not match with the flat inverts and variable internal geometry (e.g. sightline widening) required for the Project. Using a TBM would therefore result in significant over-excavation and C&D materials.

- Large-diameter TBM typically achieves low production rates of around 2m/day in hard rock stratum. By comparison, D&B performs very efficiently in hard rock and can achieve advance rates of larger than 5m per blast under favourable conditions.

2.7.5.3 D&B method is a cyclic operation comprising activities of drilling - blasting – mucking – installing support.

2.7.5.4 The supply of explosives for blasting relies on a daily delivery from Mines Division. This delivery is typically made between 11am and 2pm each day, and the explosives are expected to be used immediately, limiting to a single blast each day during the afternoon. In order to expedite the tunnel excavation with minimised idle time in the blast cycle and mitigate the potential risk of missing out the explosives delivery by Mines Division due to unforeseen site conditions e.g. extra time for temporary support, machine malfunction, an explosive magazine is required to allow flexible blasting time.

2.7.5.5 During blasting operations, the blast door at the tunnel portal will be closed to contain blasting activities within the tunnel and separate them from the surrounding environment, thereby effectively controlling potential air quality and noise impacts.

2.7.5.6 When works are undertaken near environmentally or geotechnically sensitive areas (e.g., weak ground conditions or existing water tunnels), the charge weight per blast will be carefully calculated and estimated to limit ground vibration and associated environmental impacts. Mechanical breaking shall be adopted where necessary.

2.7.5.7 Where the tunnel alignment traverses poor ground conditions or water-bearing strata, pre-excavation grouting will generally be implemented. This measure effectively controls groundwater inflow and minimises groundwater drawdown, ground settlement, and related environmental impacts.

2.7.5.8 Given its lower capital cost, shorter mobilisation period, faster excavation rate, reduced generation of C&D materials, lower environmental impacts, as well as greater flexibility and constructability, the D&B method is considered appropriate and suitable for the Project. A shortened construction programme by D&B method is not only financially advantageous but also facilitates early project commissioning and reduces the overall duration of construction-related environmental impacts.

2.7.6 Construction of Administration, Portal Plant Building / Plant Room and Satellite Control Buildings

2.7.6.1 The buildings would be constructed by conventional reinforced concrete construction method which consists of foundation construction, rebar fixing, erection of formwork and falsework and concrete casting. Precast concrete elements would be explored wherever possible to minimise on-site work and the associated environmental impacts.

2.7.7 Construction of Retaining Wall / Slope Strengthening

2.7.7.1 The site formation work for the Project would inevitably require new establishment of retaining wall and / or slope re-instatement to maintain new formed level difference and ensure stability. The type of retaining walls can be in the form of concrete, pile wall with tie back, etc. subject to the conditions of the site.

2.7.7.2 Where possible, construction programme should be arranged so that the slopes and retaining walls are constructed under favourable weather condition, preferably in dry season. Temporary drainage system shall be constructed prior to construction.

2.7.7.3 Obstructions to construction shall be verified and determined on site. Site clearance and utilities diversion will be conducted at the onset of the construction. Minimal disturbance to the environment shall be maintained during the construction.

2.7.7.4 Soil nailing is typically employed where the stability of existing slopes is compromised, for instance due to the application of additional surcharge loads or the formation of steeper slopes. Soil nailing mobilises the inherent shear strength of the in-situ soil mass for resistance, and can minimise disturbance to the surrounding environment owing to its relatively small excavation volume and footprint compared with alternative stabilisation schemes. While soil nailing is cost-effective, it has inherent limitations in terms of load-bearing and stability enhancement. Its applicability is also constrained by factors including soil conditions, groundwater levels, etc. Reinforced concrete retaining walls are generally more widely adopted, as they are suitable for most site conditions. However, where the retained height difference is substantial, the use of such walls in combination with

tie-back anchors can reduce the extent of earth formation works and minimise potential encroachments into ecologically sensitive areas.

2.8 Project Programme

2.8.1.1 The Project is tentatively scheduled to be ready for tender invitation in 2027. Subject to the further planning and actual implementation of the Project, advance work and related study might commence in the same year or in 2028.

2.8.1.2 For the purposes of Environmental Impact Assessment study, construction of the Project is anticipated to commence tentatively in 2027 with a target to commission in or before 2036. The below phase division is based on the grouping of major construction activities for environmental impact assessment purposes. In practice, due to the complexity of the Project, certain works across different phases may overlap or be carried out consecutively across phases. The actual construction sequence and schedule will be further optimised based on detailed engineering design.

Table 2.4 Tentative Construction Programme

Phase	Tentative Construction Programme	Key Activity Description
Phase 1	Q3 2027 to Q4 2030	Construction advance works and initial civil engineering, including site clearance, at-grade road site formation (comprising temporary haul roads and general site formation), and portal site formation. Later in this phase, main construction works will include like at-grade road construction (including road surfacing), viaduct construction (involving pier foundation excavation), and tunnel excavation etc.
Phase 2	Q1 2031 to Q4 2033	Full scale execution of main civil works which covers the construction of at-grade roads, viaduct construction, tunnel excavation, construction of ancillary structures, etc (e.g., administration buildings, portal plant buildings/ plant room etc.)
Phase 3	Q1 2034 to Q4 2036	Completion of all remaining construction works, including but not limited to finishing of at-grade roads, viaducts, tunnels, and ancillary buildings. It also encompasses the installation, commissioning and testing of associated electrical and mechanical (E&M) systems, as well as site reinstatement works, etc.

2.9 Concurrent Projects

2.9.1.1 In assessing cumulative environmental impacts, concurrent projects in the vicinity have been identified and reviewed, as shown in **Figure 2.3**. The status of these concurrent projects is based on the available information obtained during the course of the EIA Study. It should be noted that the implementation of individual project would be subject to further development and subsequent actions of the respective project proponent.

Table 2.5 Concurrent Projects

Project No.	Project Proponent	Project Name	Tentative Construction Programme (Reference)	Brief Description of the Project	Potential Cumulative Impact	
					Construction	Operation
1	CEDD	Kwu Tung North New Development Area	Commencement: 2017 Completion: 2031 (Approved EIA Report on North East New Territories New Development Areas (AEIAR-175/2013))	The Kwu Tung North New Development Area, occupying about 447 ha, is proposed to be developed as a 'Mixed Development Node' with a mix of residential, commercial, research and development (R&D) and agricultural uses as well as retail and services, community and government facilities and land for natural and ecological conservation.	• Air Quality • Noise • Water Quality • Ecology • Landscape	• Air Quality • Noise • Water Quality • Landscape and Visual
2	WSD	Ngau Tam Mei Water Treatment Works Extension	Commencement: 2025 Completion: 2030 (Approved EIA Report on Ngau Tam Mei Water Treatment Works Extension (AEIAR-262/2024))	The project will include the extension of NTMWTW within the site compound with the aim of upgrading the water treatment capability and enhancing the treatment efficiency. It also includes the construction of extension of NTM Fresh Water Primary Service Reservoir (FWPSR) adjacent to the existing FWPSR and a tunnel in connection with the NTMWTW, laying of fresh water trunk mains from the extended NTMWTW to near Tan Kwai Tsuen.	• Air Quality • Noise • Water Quality • Ecology • Fisheries	• Water Quality • Noise • Visual
3	CEDD	Ngau Tam Mei New Development Area	Commencement: 2027 Completion: 2036 (Approved EIA Report on Development at Ngau	Situated to the south of STT and to the northeast of Yuen Long New Town, the Project aims to make use of the existing brownfield clusters and the "Green Belt" in Ngau Tam Mei (NTM) for comprehensive development, including a university town, integrated hospital,	• Air Quality • Noise • Water Quality • Ecology • Fisheries • Landscape	• Air Quality • Noise • Water Quality • Landscape and Visual

Project No.	Project Proponent	Project Name	Tentative Construction Programme (Reference)	Brief Description of the Project	Potential Cumulative Impact	
					Construction	Operation
			Tam Mei Area (AEIAR-271/2025))	residential development, etc., leveraging the development opportunities brought by the NOL Main Line, with the support of new and upgraded infrastructures proposed at the NTM.		
4	HyD	Northern Metropolis Highway (NMH) – Kwu Tung Section	Commencement: Under Review Completion: Target completion in 3-4 years after NMH-San Tin Section commissioning (Project Profile (PP-690/2025))	The Project is to construct an approximately 5 km long dual three-lane carriageway from Fanling Highway (Pak Shek Au Section) towards the Man Kam To Area.	• Air Quality • Noise • Water Quality • Ecology • Fisheries • Landscape	• Air Quality • Noise • Water Quality • Landscape and Visual
5	HyD	Northern Metropolis Highway (NMH) – Tin Shui Wai Section	Commencement: Under Review Completion: Target completion in 3-4 years after NMH-San Tin Section commissioning (Project Profile (PP-691/2025))	The Project is to construct an approximately 5 km long dual three-lane carriageway from Tin Wah Road towards the San Tin Section of the NMH.	• Air Quality • Noise • Water Quality • Ecology • Fisheries • Landscape	• Air Quality • Noise • Water Quality • Landscape and Visual
6	MTR Corporation Limited	Northern Link (NOL)	Commencement: 2025 Completion: 2034	The proposed NOL Main Line includes an underground railway corridor of about 10.7 kilometres between the Kam Sheung Road Station and Kwu Tung Station, with three intermediate stations	• Air Quality • Noise • Water Quality • Ecology • Fisheries	• Noise • Water Quality • Landscape and Visual

Project No.	Project Proponent	Project Name	Tentative Construction Programme (Reference)	Brief Description of the Project	Potential Cumulative Impact	
					Construction	Operation
			(Approved EIA Report on Northern Link (AEIAR-259/2024))	at San Tin, NTM and Au Tau, NTM Depot and ancillary buildings.	• Landscape	
7	Infinity View Limited and Planet Universal Limited	Planned Residential Development under Planning Application no. Y/YL-NSW/7	Commencement: 2026 Completion: 2030 (Project Profile (PP-695/2025))	This Project is to construct a comprehensive residential development with wetland restoration area at various lots in D.D. 104 and adjoining government land near Wing Kei Tsuen, Yuen Long	• Air Quality • Noise • Water Quality • Landscape	• Air quality • Noise • Water Quality • Landscape and Visual
8	King Garden Limited	Planned Residential Development under Planning Application no. Y/YL-NSW/8	Commencement: No Publicly available information Completion: 2031 (Approved Planning Application No. Y/YL-NSW/8)	This Project is to construct a residential development and the associated facilities (e.g., clubhouses and carpark), as well as GIC facilities (e.g., home for the aged and elderly centre) at various lots in D.D.107 and adjoining government land, west of Castle Peak Road – Tam Mi, Yuen Long.	• Air Quality • Noise • Water Quality • Landscape	• Air quality • Noise • Water Quality • Landscape and Visual
9	Bright Strong Limited	Planned Residential Development under Planning Application no. Y/YL-NSW/9	Commencement: No publicly available information Completion: 2031 (Approved Planning Application No. Y/YL-NSW/9)	The Project is to construct a residential development and associated facilities (e.g., clubhouse, carpark), commercial podium and a transport lay-by, a school, and relocation of soy sauce factory at lots 1910 RP (Part) and 1743 S.C PR (Part) in D.D.107 and adjoining government land, west of Castle Peak Road – Tam Mi, Yuen Long.	• Air Quality • Noise • Water Quality • Landscape	• Air quality • Noise • Water Quality • Landscape and Visual
10	Topwood Limited, Success King Limited	Planned Residential Development under Land	Commencement: No publicity available information	The Project is to construct both public and private residential development, retail facilities, GIC facilities, and transport terminus at various lots in	• Air Quality • Noise • Water Quality	• Air quality • Noise • Water Quality • Visual

Project No.	Project Proponent	Project Name	Tentative Construction Programme (Reference)	Brief Description of the Project	Potential Cumulative Impact	
					Construction	Operation
	and Richduty Development Limited	Sharing Pilot Scheme Application No. LSPS/002	Completion: 2031 (Final report submitted under the Land Sharing Pilot Scheme Application No. LSPS/002)	D.D.115 and adjoining government land, Tung Shing Lei, Yuen Long.		
11	Richduty Development Limited	Planned Residential Development under Planning Application no. A/YL-NSW/274	Commencement: No publicity available information Completion: No latest publicity available information	The Project is to develop a residential and community hub (including community facilities, commercial uses, and a transport lay-by) in "Undetermined" Zone at Tung Shing Lei, Yuen Long	• Air Quality • Noise • Water Quality	• Air Quality • Water Quality • Visual
12	Fruit Design and Build Limited	Planned Residential Development under Planning Application no. A/YL-NSW/314	Commencement: No publicity available information Completion: 2027 (RNTPC Paper No. A/YL-NSW/314B)	The Project is to construct a comprehensive residential development scheme at various lots in D.D.104, North of Kam Pok Road East, Pok Wai, Yuen Long	• Air Quality • Noise • Water Quality	• Air Quality • Water Quality • Visual
13	CEDD	San Tin Technopole	Commencement: 2024 Completion: 2039 (Approved EIA Report on San Tin / Lok Ma Chau Development Node (AEIAR-261/2024))	The project covers a development area of 610 ha which is located at the heart of the NM and in proximity to Shenzhen's I&T Zone. San Tin Technopole will bolster I&T, residential spaces, community and commercial facilities, and other vital infrastructures in Hong Kong.	• Air Quality • Noise • Water Quality • Ecology • Fisheries • Landscape	• Air Quality • Noise • Water Quality • Landscape and Visual

Project No.	Project Proponent	Project Name	Tentative Construction Programme (Reference)	Brief Description of the Project	Potential Cumulative Impact	
					Construction	Operation
14	CEDD	Site Formation and Infrastructure Works for Public Housing Development at Sha Po, Shap Pat Heung and Tai Kei Leng Yuen Long	Commencement: Under review by the Government Completion: Under review by the Government	The project scope comprises site formation works, geotechnical works, roadworks, sewerage and drainage works; associated tree felling, transplanting and preservation works; and ancillary works including waterworks, landscaping works, environmental mitigation measures, etc. for public housing development at Sha Po	• Air Quality • Noise • Water Quality • Landscape	• Air quality • Noise • Water Quality • Landscape and Visual
15	AFCD	Proposed Nam Sang Wai Wetland Conservation Park	Commencement: Under review by the Government Completion: Under review by the Government	The Project is to develop wetland conservation park system to conserve the Deep Bay wetlands with ecological value and creating environmental capacity for the Northern Metropolis to achieve co-existence of conservation and development.	• Air Quality • Noise • Water Quality • Ecology • Landscape	• Water Quality • Ecology • Landscape and Visual
16	Top Field International Limited and Ideal Ace International Limited	Planned Residential Development Under the Planning Application no. A/YL-NSW/293	Commencement: No publicity available information Completion: No latest publicity available information	The project is to develop a residential development at the various lots in D.D.103 and D.D.115 in Tung Shing Lei, Yuen Long.	• Air Quality • Noise • Water Quality • Landscape	• Air quality • Noise • Water Quality • Landscape and Visual
17	Ease Gold Development Limited	Planned Residential Development Under the Planning	Commencement: No publicity available information Completion: No publicity available information	The project is to develop a residential development at lot 1071 in D.D. 103, Ha Ko Po Tsuen, Kam Tin, Yuen Long.	• Air Quality • Noise • Water Quality	• Air quality • Noise • Water Quality • Visual

Project No.	Project Proponent	Project Name	Tentative Construction Programme (Reference)	Brief Description of the Project	Potential Cumulative Impact	
					Construction	Operation
		Application no. A/YL-KTN/964				
18	Bright Strong Limited	Planned Residential Development Under the Planning Application no. A/YL-KTN/663	Commencement: No publicity available information Completion: Phase 1: Completed Phase 2: No latest publicity available information (Approved Planning Application No. A/YL-KTN/663)	The project is to develop a residential development and public vehicle park at the various lots in D.D. 107 and Adjoining Government Land, Kam Tin, Yuen Long.	• Air Quality • Noise • Water Quality • Landscape	• Air Quality • Water Quality • Landscape and Visual
19	Main Start Limited, Universal Faith Development Limited and Right Top Limited	Planned Social Welfare Facilities under the Planning Application no. A/YL-NSW-348 and A/YL-NSW-349	Commencement: No publicity available information Completion: 2030 (Planning Application Nos. A/YL-NSW/348 and A/YL-NSW/349)	The Project is to develop a 3-storey Residential Care Home for the Disabilities (RCHD) in various lots in D.D. 104, Nam Sang Wai.	• Air Quality • Noise • Water Quality	• Air quality • Noise • Water Quality • Visual
20	Global King Investment Limited, Winpost (HK) Investment	Planned School under the Planning	Commencement: No publicity available information	The Project is to build and operate an international school which is located at various lots in D.D.94, D.D.98 and D.D100 and adjoining Government	• Air Quality • Noise • Water Quality	• Air quality • Noise • Water Quality • Visual

Project No.	Project Proponent	Project Name	Tentative Construction Programme (Reference)	Brief Description of the Project	Potential Cumulative Impact	
					Construction	Operation
	Limited and Rand Development Limited	Application no. Y/NE-KTS/18	Completion: 2036 (Planning Application No. Y/NE-KTS/18)	Land, in Kwu Tung South in the New Territories.		
21	Bright Strong Limited	Planned Residential Development Under the Planning Application no. A/YL-KTN/604	Commencement: No publicity available information Completion: Phase A: No latest publicity available information Phase B: No publicity available information, subject to land acquisition progress. (Approved Planning Application No. A/YL-KTN/604)	The project is to develop a residential development and public vehicle park at the various lots in D.D. 107 and Adjoining Government Land, Kam Tin, Yuen Long.	• Air Quality • Noise • Water Quality • Landscape	• Air Quality • Water Quality • Landscape and Visual

2.10 Key Public Concerns

2.10.1.1 During the public inspection of the Project Profiles (No. PP-683/2025) held from 23 April 2025 to 6 May 2025, several public comments were received. Public comments primarily focus on ecological impacts, water quality, cumulative impacts and alternative alignments.

2.10.1.2 The Project design has taken into consideration the public views and concerns, and mitigation measures have been recommended as appropriate to minimise potential environmental impacts. The key views / concerns and the corresponding design considerations and follow up actions taken are summarised.

Table 2.6 Summary of Key Public Concerns and Follow-up Actions Taken

Item No.	Key Concerns	Follow up Actions Taken
1.	Potential ecological impacts due to habitat loss and habitat fragmentation, isolation, roadkill risk and glare.	• The design of the NMH (San Tin Section) has considered the habitat loss, habitat fragmentation, isolation, road kill and glare impact. Please refer to **Section 2.4** for the detailed design of the alignment and at grade/viaduct tunnel sections. • Conducted 12-month ecological surveys covering both wet and dry seasons within the 500m assessment area, and supported by literature review (e.g., The State of Hong Kong Biodiversity 2025, etc.) to establish the ecological profile of the Project, and implement avoidance and minimisation measures, such as erecting fences at the at-grade road sections near Long Ha to minimise roadkill, adopt intermittent at-grade and viaduct design for the alignment near LTCP to minimise habitat fragmentation and isolation and facilitate wildlife movement, adopt viaduct with limited number of piers at the wetland habitat to the west of NTMWTW to minimise habitat loss, and on-site habitat compensation to enhance ecological connectivity. • Use of directional lighting systems to avoid light spill to sensitive areas (e.g., LTCP) so as to minimize glare impacts on nocturnal species. • Please refer to **Section 8** (Ecological Impact).
2.	Potential groundwater drawdown affecting fishponds and reservoirs, pollution from untreated surface runoff, river connectivity and flow of watercourses feeding into Sheung Yue	• Adopt groundwater control strategies to minimize groundwater infiltration (e.g. sealing up groundwater carrying rock joints and weak zones, to limit

Item No.	Key Concerns	Follow up Actions Taken
	River and Long Valley Nature Park and impacts on water quality.	the groundwater ingress into the tunnel during tunnel construction) • Implement appropriate mitigation measures and good site practices to control construction site runoff and avoid adverse water quality impact to the watercourses and other water sensitive receivers. • There is no tunnel construction underneath fishponds and reservoirs for this Project. Potential groundwater drawdown affecting fishponds and reservoirs is not anticipated • Please refer to **Section 5** (Water Quality Impact) & **Section 8** (Ecological Impact) • With implementation of the proven groundwater control measures, no adverse water quality impact from groundwater drawdown is expected.
3.	Compliance with Air Quality Objectives (AQOs) and noise criteria.	• Air quality and noise impact assessments for both construction and operation phases have been conducted. • Implementation of dust suppression measures and use of quieter equipment during construction have been proposed. • Meanwhile, noise mitigation measures such as Low Noise Road Surfacing (LNRS) and noise barriers are proposed to reduce noise impacts arising from traffic during operation phase, particularly to protect the planned residential development nearby the existing San Tin Highway. • Please refer to **Section 3** (Air Quality) & **Section 4** (Noise) for the assessment findings.
4.	Impacts on Ngau Tam Mei Site of Archaeological Interest.	• The Archaeological Impact Assessment for the Ngau Tam Mei Site of Archaeological Interest (SAI) has been conducted. No sub-soil disturbance works would be carried out within the Ngau Tam Mei SAI. Impact on Ngau Tam Mei SAI is not expected. • Please refer to **Section 11** for Cultural Heritage Impact details.
5.	Disposal of Construction and Demolition (C&D) Materials	• Implementation of a trip ticket system to avoid illegal dumping and control fly tipping. • Adopt good site practices such as to minimize the potential environmental

Item No.	Key Concerns	Follow up Actions Taken
		impact associated with generation and disposal of C&D materials. • Preparation of a Waste Management Plan as part of the Environmental Management Plan to monitor and audit disposal of C&D materials. • Adopt real time tracing and monitoring of their travel routings and parking locations of the dump trucks through GPS or equivalent system, would avoid / minimise the chance of illegal dumping. • Please refer to **Section 6** for Waste Management Implication
6.	Potential high arsenic content in soil	• A separate arsenic assessment and remediation plan should be prepared, in consultation with EPD, and submitted to EPD for endorsement prior to the commencement of construction works. • Please refer to **Section 7** (Land Contamination)
7.	Alternatives for construction of tunnel and habitat loss at tunnel portal	• TBM is not adopted having considered capacity of typical hard rocks TBM and the Project's need as well as cost effectiveness. Moreover, tunnel section (circular shape) created by TBM would generate more C&D materials than section created by drill and blast method (horseshoe). • The environmental impacts of the D&B methods have been evaluated in the EIA study, with implementation of mitigation measures, adverse impact is not anticipated. • Furthermore, slope cutting has been minimised to reduce habitat loss at tunnel portal. Retaining walls / soils nail or combination of both may be proposed to allow for steeper slope gradients and thus reduce the extent of slope excavation. • Please refer to **Section 2.4** and **2.7** for details.
8.	Cumulative environmental impacts from adjacent developments.	• Assessed cumulative impacts during both construction and operation phases including the overlapping programmes and phased Implementation of NMH sections in EIA and include all potential environmental impacts arising from the concurrent projects within the assessment area.

Item No.	Key Concerns	Follow up Actions Taken
9.	Environmental Impacts to the adjacent planned developments	• Identified these developments as sensitive receivers within the assessment area and conducted in-depth assessment. • Appropriate noise mitigation measures such as Low Noise Road Surfacing (LNRS) and noise barriers are proposed to reduce traffic noise impacts during operation phase, particularly to protect the planned residential development adjacent to the Project. • Stakeholder engagement and regular updates to the affected parties are required.
10.	Phased implementation and EIA submission of the Northern Metropolis Highway	• Adopt the strategy of "holistic planning and staged implementation" for NMH. • Reviewed the traffic need, implementation programme of other infrastructure projects, cost effectiveness, and public resources, etc. to formulate the implementation programme for each section of the NMH. • The EIA includes assessment of cumulative impact of the interconnected NMH sections, including the phased implementation scenario, to ensure no underestimated of overall environmental impacts.
11.	Associated Off-Site and Secondary Environmental Impacts	• Evaluate indirect and off-site impacts within the assessment area. • Implementation of a trip ticket system to avoid illegal dumping and control fly tipping. • Adopt real time tracing and monitoring of their travel routings and parking locations of the dump trucks through GPS or equivalent system, would avoid / minimise the chance of illegal dumping. • Please refer to **Section 3** (Air Quality), **Section 4** (Noise), **Section 5** (Water Quality), **Section 8** (Ecology), **Section 9** (Fisheries), **Section 10** (Landscape and Visual Impact) for the associated off-site and secondary environmental impact within their respective assessment areas.
12.	Alternative alignment options near Pok Wai and LTCP	• The current alignment has considered perspectives of site

Item No.	Key Concerns	Follow up Actions Taken
		constraints, traffic performance, engineering constructability, social acceptance, environmentally friendliness, etc. • Current alignment has minimised encroachment into ecologically sensitive areas such as gazetted Conservation Area, and avoided direct encroachment to LTCP, existing and planned bat roost at Pok Wai, and natural watercourses near Pok Wai Public School. • The alignment has been carefully designed from a road safety perspective while maintaining sufficient clearance from the future Pok Wai Ancillary Building and the existing CLP pylon. • Aesthetically pleasing design of aboveground structures shall be adopted to blend in the structures to the adjacent landscape and visual context. • Please refer to Section 2.6 – Consideration of Alignment Option and **Section 10** (Landscape and Visual Impact). • Viaduct was adopted as current alignment at the NMH San Tin Section I/C 02 area, as compared to tunnel scheme, the current at-grade and viaduct design has enhanced connectivity to Tsing Long Highway and New Development Area (NDA), better traffic performance and engineering merits including less construction risks. • Adoption of tunnel scheme locally at Pok Wai Area was reviewed, but compared to current viaduct and at-grade road alignment, locally adopting tunnel section at Pok Wai area would require extensive tunnel portal and the associated earthworks at Kai Kung Shan and would impose larger impact on CA Zone and encroach into Lam Tsuen Country Park.